IERIN P MENACHERY

AI & Machine Learning | LLMs & RAG | Software Development

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PROFESSIONAL SUMMARY

Computer Science undergraduate with hands-on experience in Machine Learning, Artificial Intelligence, LLMs, and Retrieval-Augmented Generation (RAG). Experienced in data preprocessing, feature engineering, model training and evaluation, and developing AI-driven applications using Python and machine learning frameworks. Developed Hydrology GPT, an LLM- and RAG-based domain-specific question-answering system during an AI/ML internship at Larsen & Toubro (L&T). Seeking entry-level opportunities in Machine Learning, AI, or related software engineering roles.

TECHNICAL SKILLS

Languages	Java, C, C++, SQL, Python, HTML, CSS
Artificial Intelligence & Machine Learning	Machine Learning, Predictive Modeling, Linear Regression, Logistic Regression, Artificial Neural Networks (ANN), Data Preprocessing, Feature Engineering, Model Training, Model Evaluation
Generative AI & LLMs	Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Prompt Engineering, Question Answering Systems, Embeddings, Information Retrieval, Domain-Specific AI Applications
Mobile / Android	Android Studio, Java for Android, XML Layouts, SQLite, Activity & Navigation
Libraries & Frameworks	NumPy, Pandas, Scikit-learn, PyTorch, Matplotlib
Tools & Platforms	GitHub, VS Code, Android Studio, Linux, MySQL
Concepts	OOP, Data Structures, DBMS
Soft Skills	Effective Communication, Problem-Solving, Leadership, Teamwork, Organization

WORK EXPERIENCE

Artificial Intelligence & Machine Learning Intern — Larsen & Toubro (L&T) *30 Jun 2026 – 29 Aug 2026*

Mumbai, India

- Developed Hydrology GPT, a domain-specific LLM & RAG system, and applied ML models to flood and environmental data using python.
- Presented Hydrology GPT to L&T's Hydrology and Structural Departments, with teams from Delhi and Chennai joining remotely.

Python & Single Board Computers Intern — CSEED *16 Jun 2025 – 22 Jun 2025*

Kerala, India

- Explored hardware-software integration through guided lab exercises, gaining exposure to embedded systems and IoT prototyping concepts.

Android App Development Intern — ICT Academy of Kerala *Jul 2024 – Aug 2024*

Kerala, India

- Developed a Java-based Android course management app and gained hands-on experience in Android Studio, Git, UI design, debugging, and APK testing.

PROJECTS

Hydrology GPT — AI-Powered Hydrology & Flood Question-Answering System *2026*

- Developed a domain-specific question-answering system using LLMs and RAG, integrating hydrology knowledge for context-aware and grounded responses.

- Built a retrieval pipeline using domain-specific hydrology knowledge, integrating relevant information as context for improved response grounding.
- Presented the system to L&T's Hydrology and Structural departments, with teams from Delhi and Chennai joining remotely.

Blockchain-based Chain of Custody System *2026*

- Developed a blockchain-based system using SHA-256 hashing to ensure data integrity, detect unauthorized modifications, and maintain a verifiable audit trail.

Fraud Detection Tool for Non-Technical Users *2026*

- Developed an intuitive and accessible system to identify fraudulent URLs, phishing messages, and scam communications, with a focus on usability for elderly and non-technical users.

Smart Intravenous Fluid Monitoring System *2024*

- Designed a low-cost IoT-based IV fluid monitoring system using a load-cell sensor, microcontroller, and alert mechanism; presented the concept at KSCSTE Tech Fest, IIT Palakkad (2024).

Course Management Android App *2024*

- Built an Android application for course browsing, details, and local favorites, using multiple Activities, Intent-based navigation, and XML layouts

EDUCATION

B.Tech in Computer Science (Third Year) *Aug 2023 – Aug 2027*
Jyothi Engineering College, Thrissur, Kerala **CGPA: 7.97 / 10**

Higher Secondary (Computer Science Stream) *2020 – 2022*
Sacred Heart C.G.H.S., Thrissur, Kerala **96.3%**

Diploma in Computer Application *2020 – 2022*
SCOLE Kerala, Thrissur, Kerala **Grade A+**

ACHIEVEMENTS & CO-CURRICULAR

Academic & Technical

- **State A Grade** — IT Presentation, IT Mela (Kerala State)

Sports & Adventure

- **District 1st** — Fencing & Mountaineering

Leadership & Arts

- **Rajya Puraskar Awardee** — Received the Rajya Puraskar from Bharat Scouts & Guides with the Nanmamudra Unit.
- **State 2nd** — Monoact (Solo Acting) competition conducted by Malayala Manorama

CERTIFICATIONS

- **Sensors and Actuators** — NPTEL, IISc Bangalore — Score: 61%
- **Embedded Systems Design** — NPTEL, IIT Kharagpur
- **Web Development with AI** — Internshala Trainings — Final Assessment: 81%
- **Artificial Intelligence for All** — Infosys Springboard (2025)
- **5-Day AI Agents Intensive Course with Google** — 2025
- **Basics of Python** — Infosys Springboard (2026)
- **Introduction to Modern AI** — Cisco Networking Academy (2025)
- **Introduction to Data Science** — Cisco Networking Academy
- **Introduction to Cybersecurity** — Cisco Networking Academy

REFEREES

Dr. Vamsi Krishna Gunturu
Assistant Engineering Manager, Larsen & Toubro
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Ms. Sudesna Biswas
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